Small Engines Overhaul · Unit 3 · Level 2 (modified)

4-Stroke Theory & Engine Parts



TODAY'S PLAN

- ⚙️ The four strokes
- 🔩 Main engine parts
- 🔁 How motion is made
- 2️⃣ 2-stroke vs 4-stroke

Review · 5 minutes

Quick Review



Name two engine careers.

What is an apprenticeship?

Name one soft skill.

Words to Know

Words to Know



Intake

Stroke 1: the piston moves down and the intake valve opens to pull in the air-fuel mix.



Compression

Stroke 2: the piston moves up with both valves closed and squeezes the mix.



Power

Stroke 3: the spark plug fires, the mix burns and pushes the piston down.



Exhaust

Stroke 4: the piston moves up and the exhaust valve opens to push burned gas out.

Words to Know

More Words to Know



🔩 **Piston**: The round part that slides up and down in the cylinder.

🔁 **Crankshaft**: The shaft that changes the piston's up-and-down motion into spinning motion.

🚪 **Valve**: A part that opens and closes to let the mix in or exhaust out.

🛞 **Flywheel**: A heavy wheel on the crankshaft that keeps the engine turning smoothly and holds the magnets for the spark.

Learn

The Four Strokes



Intake

piston down, intake valve open



Compression

piston up, valves closed



Power

spark, piston pushed down



Exhaust

piston up, exhaust valve open

🃏 Check 1 of 2



During compression, the valves are…

Hold up 1, 2, or 3:

1️⃣ Both open

2️⃣ Both closed

3️⃣ Only intake open

✅ Answer · Card 1 of 2



Both closed

Closed so the mix can be squeezed.

Learn

One Power Stroke Every 2 Turns



The crankshaft turns **2 times** for the 4 strokes.

Only the **power** stroke pushes; the flywheel carries the rest.

The **camshaft** opens the valves at the right time.

Learn

Main Parts



Piston + rings
seal and move



Connecting rod
links piston to crank



Crankshaft
makes spinning motion



Valves + camshaft
let gas in and out

Learn

Outside Parts

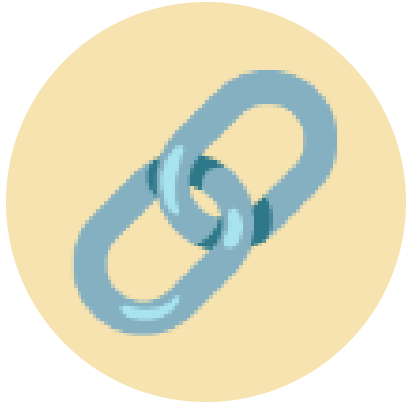


Flywheel: smooths turning; holds magnets for ignition.

Carburetor: mixes air and fuel; **air filter** cleans the air.

Spark plug, **muffler**, **governor** (controls speed).

🃏 Check 2 of 2



What links the piston to the crankshaft?

Hold up 1, 2, or 3:

1️⃣ Spark plug

2️⃣ Muffler

3️⃣ Connecting rod

✅ Answer · Card 2 of 2



Connecting rod

The rod.

Compare

2-Stroke vs 4-Stroke



2️⃣ 2 -stroke

Power every turn
Oil mixed in the fuel
Chainsaws, trimmers, outboards

4️⃣ 4 -stroke

Power every 2 turns
Oil in the crankcase
Mowers, cars, generators

Wrap-Up

Recap



4 strokes
intake, compression, power, exhaust



Piston
up and down



Crankshaft
spinning



2-stroke
oil in fuel

Wrap-Up

Exit Ticket



List the 4 strokes in order.

Which part turns up-and-down into spinning?

One difference between 2-stroke and 4-stroke?